



Original article

Drug repurposing? Cardiovascular effect of colchicine on patients with coronary artery disease: A systematic review and meta-analysis



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ABSTRACT

Background: Patients with coronary artery disease (CAD) are at high risk of atherosclerotic events. The aim of this meta-analysis is to evaluate the cardiovascular protective effect of colchicine on patients with CAD.

Methods: In this systematic review and meta-analysis, we searched PubMed and Embase for studies published until April 28, 2020. We included studies that reported the incidence of myocardial infarction (MI), restenosis after percutaneous coronary intervention (PCI), and mortality for CAD patients within colchicine and control (placebo or usual care) groups. A random-effects meta-analysis model was then applied.

Results: Ten eligible trials were identified, including 6398 patients (3248 received colchicine while 3150 were controls). The risk of composite events of MI and restenosis after PCI was significantly decreased with colchicine treatment [odds ratio (OR) 0.48, 95% confidence interval (CI) 0.28–0.79]. We found a similar trend of lowered risk of MI in the colchicine group, although without statistical significance (OR 0.41, 95% CI 0.16–1.08). The risk of restenosis after PCI also decreased significantly with colchicine treatment (OR 0.46, 95% CI 0.23–0.92). There was no significant difference in all-cause mortality between the two groups (OR 0.80, 95% CI 0.56–1.15). The included patients had significantly higher risks of gastrointestinal (GI) events with colchicine treatment.

Conclusions: This meta-analysis shows that there is a decreased composite risk of MI and restenosis after PCI with the use of colchicine in patients with CAD. However, colchicine did not appear beneficial for all-cause mortality, and it led to a higher risk of GI events.

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Introduction

Colchicine is an old drug usually used for the treatment of gout. It has anti-inflammatory properties by preventing the assembly of microtubules, thereby disrupting inflammatory activation, chemotaxis of inflammatory microtubule-based cells, leukotriene and cytokine generation, and phagocytosis [1]. Due to this property, colchicine has also been used as a treatment or prophylaxis for many inflammatory diseases and conditions, such as pericardi-

itis, familial Mediterranean fever, Behçet's disease, postcardiotomy syndrome, and postoperative atrial fibrillation (AF) [2–4]. In recent years, coronary artery disease (CAD) has also been found to involve an ongoing inflammatory response. Atherosclerosis, the main cause of CAD, is an inflammatory disease. Inflammatory cytokines, proteases, coagulation factors, radicals, and vasoactive molecules produced by activated macrophages, T cells, and mast cells at sites of atherosclerotic lesion destabilize the lesion and thus cause the rupture of the plaque. The rupture of the plaque results in acute thrombotic occlusion of a coronary artery [5, 6]. Accordingly, colchicine may have the ability to slow the progression of atherosclerosis, and it has been used to reduce the risk of myocardial infarction (MI) in patients with stable CAD or after acute coro-

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nary syndrome (ACS) [7, 8]. Moreover, both the anti-inflammatory and antiproliferative effects of colchicine could interfere in the formation of neointima in coronary stents [9]. Colchicine has also been used to reduce restenosis after percutaneous coronary intervention (PCI) [10]. However, the beneficial effects of colchicine remain controversial. The variations in the reported outcomes across trials may have resulted from, for example, different patient populations, doses of colchicine, or durations of follow-up. In addition, approximately 7% to 10% of patients receiving colchicine therapy experienced gastrointestinal (GI) events, especially diarrhea [11], resulting in poor compliance. Patients with a history of CAD have a higher recurrence rate of cardiovascular events [12]. Both MI and restenosis of stented coronary arteries are believed to be related to the inflammation of coronary arteries, resulting in coronary artery occlusion. Given the above-mentioned mechanisms of colchicine activity, the aim of this meta-analysis is to evaluate the cardiovascular protective effects of colchicine on patients with CAD.

Methods

Literature search

Two investigators (YYT and HKH) independently searched PubMed and Embase from their inception through April 28, 2020. The intervention group was patients with CAD receiving colchicine treatment. The control group was patients not receiving colchicine or those receiving placebo. Except for the difference in colchicine use, both the treatment and control groups received comparable and optimal care for acute cardiovascular events or stable CAD. We used the keywords “colchicine”, “coronary artery disease”, “acute coronary syndrome”, “ischemic heart disease”, “myocardial infarction”, and “percutaneous coronary intervention” for searching. The search details are shown in the Supplementary Materials (Online Table 1). The bibliographies of recent review articles, retrieved articles, and previous meta-analyses were also manually searched for relevant studies.

Study outcomes

The primary outcomes were MI, restenosis after PCI, and all-cause mortality. Considering both MI and restenosis can lead to occlusive conditions of the coronary arteries, we reasoned that colchicine has a similar effect on both outcomes. To ensure the validity of this reasoning, we performed Cochran's Q test to assess heterogeneity between these two outcomes. When there was no evidence of heterogeneity, we would pool the data together as one composite outcome. If heterogeneity was evident, MI and restenosis after PCI would be assessed separately. The secondary outcome was the occurrence of GI events in patients within the trials identified by our search strategy.

Study selection

YYT and HKH reviewed all titles and abstracts retrieved from the literature search to determine the eligibility of a study. We included human clinical trials where patients were randomly allocated to different treatment groups. Eligible studies included patients with CAD, such as stable CAD, after ACS who received treatments before or after PCI. We excluded conference proceedings without full text, non-randomized controlled trials, and studies that did not report any primary outcome of interest specific for CAD patients.

Data extraction

The outcomes (i.e. MI, restenosis after PCI, all-cause mortality, and GI events) were extracted independently from the included

studies by the two investigators mentioned above. Information on the country of study, study design, study participants, duration of follow-up, and dosage of colchicine was also collected. Discrepancies between the two reviewers were resolved through discussion and consensus.

The duration of follow-up and the dosage of colchicine were different in each trial. In the subgroup analysis, we defined long duration of follow-up as at least 6 months and short duration of follow-up as less than 6 months. At least 1 mg of colchicine per day was defined as the standard dose, and less than 1 mg per day was defined as the low dose. If the trial had different loading and maintenance doses, we used the loading dose for assessing the results.

We extracted MI data from non-stent-related MI and type 1 MI using universal definitions if the trials had reported PCI-related MI and non-PCI-related MI separately [13].

For the secondary outcome, we selected the most common GI event reported in the trial if the trial had reported two different GI symptoms.

Quality assessment of methods

We used the Cochrane Risk of Bias Tool to assess the quality and risk of bias for the included studies (Online Table 2). We defined the risk of bias as adequate (low risk of bias), unclear (undetermined risk of bias), or inadequate (high risk of bias) for assessing six aspects of the trials: sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, and selective reporting.

Statistical analysis

Odds ratios (ORs) and 95% confidence intervals (95% CIs) were calculated for the primary outcomes using the Peto odds ratio model because the event numbers were small or even zero in some studies. This method performs well when events are very rare [14]. The inverse variance method was used for pooling ORs for the secondary outcome because the event numbers were greater for this outcome. The I^2 statistic was used to evaluate heterogeneity in the studies. We defined low, moderate, and high heterogeneity as $I^2 \leq 25\%$, 25–50%, and $\geq 50\%$, respectively.

Due to different durations of follow-up and the potentially different efficacies of colchicine dosages, we performed subgroup analysis to explore the influence of these factors. If the OR of the outcome of interest was extremely high in some trials, we would exclude those trials and conduct a sensitivity analysis to evaluate the influence of these trials.

We used the statistical software package R, version 3.5.2, to analyze the data. The meta-analysis was conducted and reported following the recommendations of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement and was registered in PROSPERO (registration number: CRD42020186372).

Results

Our literature search identified 1333 potentially eligible studies. A total of 152 were duplicates and 1181 were excluded after reviewing the titles and abstracts. Of the remaining 20 articles, three were not available as full texts, three were not randomized controlled trials, three did not report any outcome of interest, and one did not report data specific for CAD patients [15]. Ten randomized controlled trials were finally included in our systematic review, totaling 6699 people with CAD [7–10,16–21] (Fig. 1). Table 1 outlines the basic characteristics of the included studies. The included

Table 1.
Overview of included studies.

References	Countries	Study design	Patient population	Duration of Follow-up	Colchicine dose	No of patients (Colchicine/Control)	Primary endpoint	Secondary endpoint
O'Keefe [10]	USA	RCT, double-blind	Elective coronary angioplasty	6 months	0.6 mg BID	197 (130/67)	Angiographic restenosis	-
Nidorf (LoDoCo) [7]	Australia	RCT, observer-blind	Stable CAD	36 months (median)	0.5 mg QD	532 (282/250)	Composite of ACS, fatal or nonfatal out-of-hospital cardiac arrest, or noncardioembolic ischemic stroke	Individual components of the primary outcome; the components of ACS unrelated to stent disease
Deftereos [9]	Greece	RCT, double-blind	DM for PCI	6 months	0.5 mg BID	196 (100/96)	Angio-ISR and IVUS-ISR	Late lumen loss; lumen area loss; percentage of neointima volume
Deftereos [16]	Greece	RCT, double-blind	STEMI, treated with PCI	5 days	Colchicine Loading: 2 mg and continuing with 0.5 mg BID	151 (77/74)	Area under the curve of CK-MB concentration; absolute myocardial infarct volume	Myocardial infarct volume; relative infarct size
Zarpelon [17]	Brazil	RCT, open-label	Elective myocardial revascularization surgery	Colchicine: 14.5 ± 11.5 days (mean)Control: 13.3 ± 9.4 days (mean)	24 hours before surgery: 1 mg, BID. Followed by 0.5 mg, BID, until discharge 1 mg QD	140 (71/69)	AF-POMR rate	Death from any cause; hospital length of stay; postoperative infection
Akodad (COLIN) [18]	France	RCT, open-label	STEMI, treated with PCI	1 month		41 (23/21)	CRP peak value during the index hospitalization	Troponin peak; tolerance of colchicine; hospitalization duration; major adverse cardiac events
Hennessy (LoDoCo-MI) [8]	Australia	RCT, double-blind	Type 1 acute MI within the prior 7 days	30 days	0.5 mg QD	237 (119/118)	Proportion of patients with a residual CRP level ≥ 2mg/L at 30 days	Actual levels of CRP at 30 days; the relative and absolute change in CRP levels from baseline to 30 days
Habib (CISR) [19]	Palestine	RCT, open-label	Stable IHD or ACS, scheduled for coronary angioplasty procedure	6 months	0.5 mg BID	60 (30/30)	Clinical ISR	Target-vessel revascularization ; stent thrombosis
Tardif (COLCOT) [20]	Canada	RCT, double-blind	Acute MI, completed any planned percutaneous revascularization procedures	22.6 months (median)	0.5 mg QD	4745 (2366/2379)	Composite of death from CV causes, resuscitated cardiac arrest, MI, stroke, or urgent hospitalization for angina leading to coronary revascularization in a time-to-event analysis	Components of the primary efficacy end point; a composite of death from CV causes, resuscitated cardiac arrest, MI, or stroke; total mortality in time-to-event analysis
Shah (COLCHICINE-PCI) [21]	USA	RCT, double-blind	IHD or ACS, treated with PCI	30 days	1.2 mg 1-2 hours before coronary angiography, following by 0.6 mg 1 hour later or immediately preprocedure	400 (206/194)	PCI-related myocardial injury	The occurrence of 30-day MACE; a composite of the earliest occurrence of death from any cause, nonfatal MI, or target vessel revascularization; PCI-related MI

BID, twice a day; QD, once a day; DM, diabetes mellitus; Angio-ISR, angiographic in-stent-restenosis; IVUS, intravascular ultrasound in-stent-restenosis; STEMI, ST elevation myocardial infarction; CK-MB, creatine kinase-myocardial band; AF-POMR, atrial fibrillation in the postoperative period of myocardial revascularization surgery; CRP, C-reactive protein; IHD, ischemic heart disease; CV, cardiovascular; MACE, major adverse cardiac events; RCT, randomized controlled trial; PCI, percutaneous coronary intervention; ACS, acute coronary syndrome; MI, myocardial infarction.

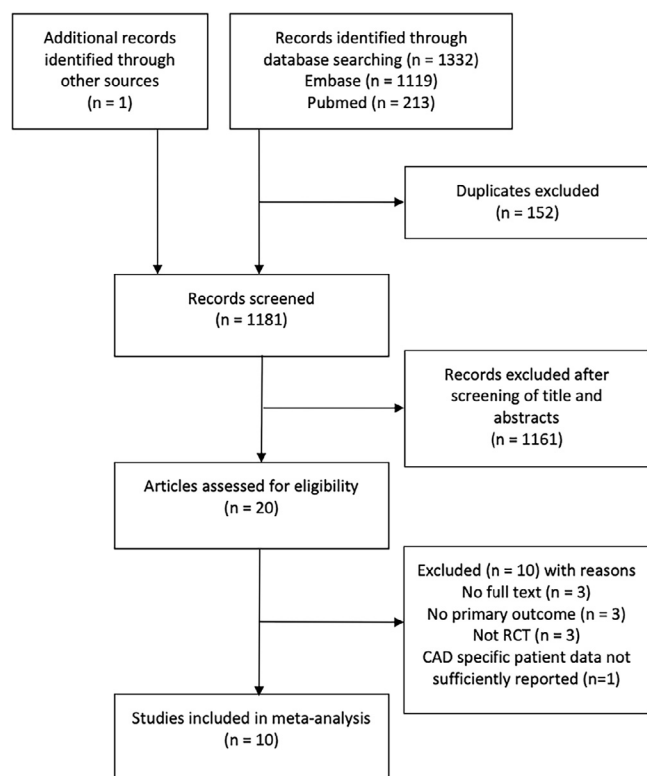


Fig. 1. Flowchart of the process to identify eligible studies with reasons for inclusion or exclusion.

CAD, coronary artery disease; RCT, randomized controlled trial.

studies were published between 1992 and 2020 and were conducted in several different countries. Six studies were double-blind, three studies were open label, and one study was observer-blind. The patient population included patients with CAD, such as stable CAD, after ACS and their status pre- or post-PCI was provided. The PCI could have been scheduled or performed urgently due to ACS. The duration of follow-up ranged from 5 days to 36 months. The colchicine doses ranged from 0.5 mg once daily to 0.6 mg twice daily. In two trials, the patients were treated with loading doses of 2 mg and 1.2 mg, respectively [16, 21].

Myocardial infarction or restenosis after percutaneous coronary intervention

According to the design before the study, we had assessed heterogeneity between MI and restenosis after PCI at the beginning of the study. The point estimation for the OR of these two outcomes was similar (OR 0.41 vs 0.46, p for Cochran's Q test = 0.85). Thus, MI and restenosis after PCI were determined as composite primary outcomes within this meta-analysis.

The risk of MI / restenosis after PCI was significantly lower in patients with the use of colchicine. Specifically, a total of 151 events out of 3248 patients was observed in the colchicine group compared to 180 events out of 3150 patients in the control group (OR 0.48, 95% CI 0.28–0.79; Fig. 2). There was high heterogeneity among the studies ($I^2=56\%$).

In the separate analysis for MI and restenosis, the results showed a nonsignificant but decreased trend in the risk of MI with colchicine. A total of 93 MI events were documented during follow-up out of 2988 patients in the colchicine group compared to 116 such events out of 2957 patients in the control group (OR 0.41, 95% CI 0.16–1.08; Fig. 2). According to the analysis of restenosis after PCI, the results showed a significant decrease in the risk of

restenosis after PCI with colchicine. A total of 58 restenosis events after PCI were documented during follow-up out of 260 patients in the colchicine group compared to 64 such events out of 193 patients in the control group (OR 0.46, 95% CI 0.23–0.92; Fig. 2). Heterogeneity in both analyses was high ($I^2=55\%$ and 52% , respectively).

We then explored heterogeneity by performing subgroup analysis according to the duration of follow-up and the dose of colchicine. In the subgroup analysis of duration of follow-up, the risk of MI / restenosis after PCI was significantly lower with the use of colchicine than without its use for both long duration and short duration of follow-up (long: OR 0.52, 95% CI 0.30–0.89; Online Fig. 1 vs short: OR 0.13, 95% CI 0.02–0.93; Online Fig. 2). The incidence of MI in the short duration of follow-up group also significantly decreased with the use of colchicine (OR 0.13, 95% CI 0.02–0.93; Online Fig. 2). The restenosis events after PCI were all within the long duration of follow-up group.

Regarding the subgroup analysis of colchicine dose, the risk of MI / restenosis after PCI significantly decreased in the standard-dose subgroup (OR 0.46, 95% CI 0.26–0.80; Online Fig. 3). However, the risk of MI or restenosis after PCI was not statistically different in the low-dose subgroup (OR 0.47, 95% CI 0.16–1.38; Online Fig. 4). All of the restenosis events after PCI occurred within the standard dose of colchicine group.

All-cause mortality

There was no difference in the risk of all-cause mortality between colchicine and placebo or no treatment. Fifty-five deaths from any cause were documented during follow-up out of 3366 patients in the colchicine group compared to 66 deaths out of 3263 patients in the control group (OR 0.8, 95% CI 0.56–1.15; Fig. 3). Nine studies reported death events; however, two of these studies showed no mortality during the entire trial period. Heterogeneity was low among the studies in this analysis ($I^2=0\%$). Causes of death is presented in supplementary material (Online Table 3).

The subgroup analyses of all-cause mortality with respect to the trial duration and colchicine dose showed no significant differences between the treatment group and control group (Online Fig. 5, Online Fig. 6).

Gastrointestinal events

A significant difference in the risk of GI events was found between colchicine and placebo or no treatment. A total of 563 GI events were documented during follow-up out of 3419 patients in the colchicine group compared to 442 GI events out of 3315 patients in the control group (OR 4.25, 95% CI 1.81–9.96; Fig. 4). The heterogeneity among these trials was high ($I^2=85\%$). There were two trials with extremely high ORs (Nidorf et al. and Akodad et al.) [7, 18]. Thus, we conducted a sensitivity analysis excluding these two trials, which also showed a significant difference in the risk of GI events between the two groups (Online Fig. 7). The heterogeneity among these trials remained high. The detail of GI events among trials is shown in supplementary material (Online Table 4).

Discussion

Our results showed that the composite incidence of MI and restenosis after PCI decreased significantly with the use of colchicine. A similar trend was found in the analysis of MI, although without statistical significance. Besides, there was a protective, beneficial effect of colchicine on the risk of restenosis after PCI. In addition, the results of all-cause mortality did not show a significant difference between the colchicine and control groups.

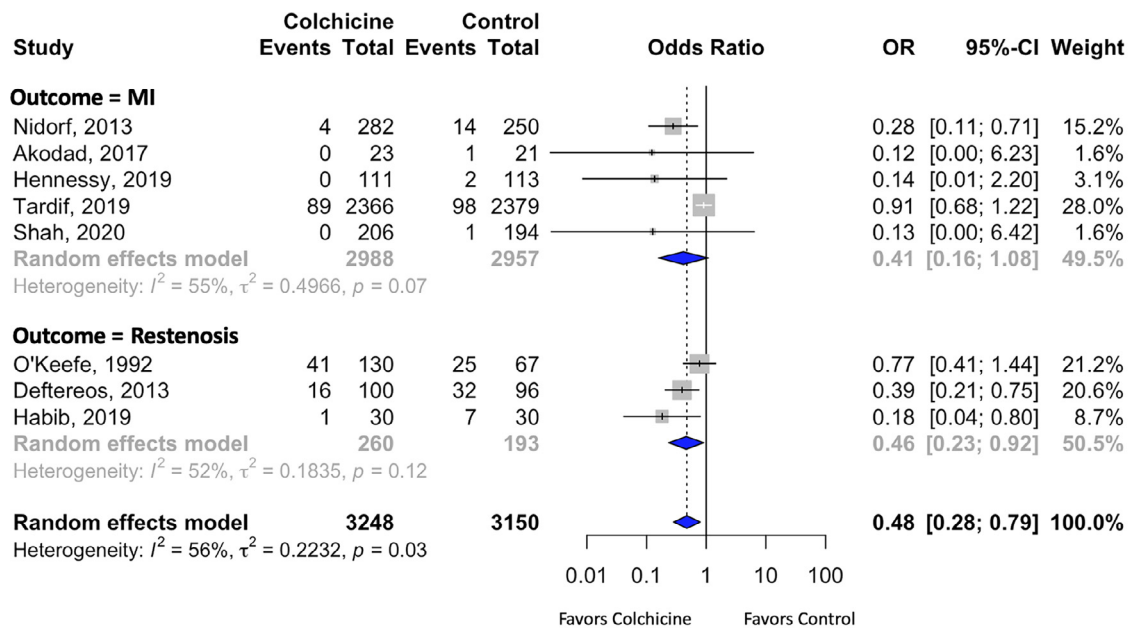


Fig. 2. Effect of colchicine on incidence of myocardial infarction, restenosis after percutaneous coronary intervention, and overall risk of myocardial infarction / restenosis after percutaneous coronary intervention.

OR, odds ratio; CI, confidence interval; MI, myocardial infarction.

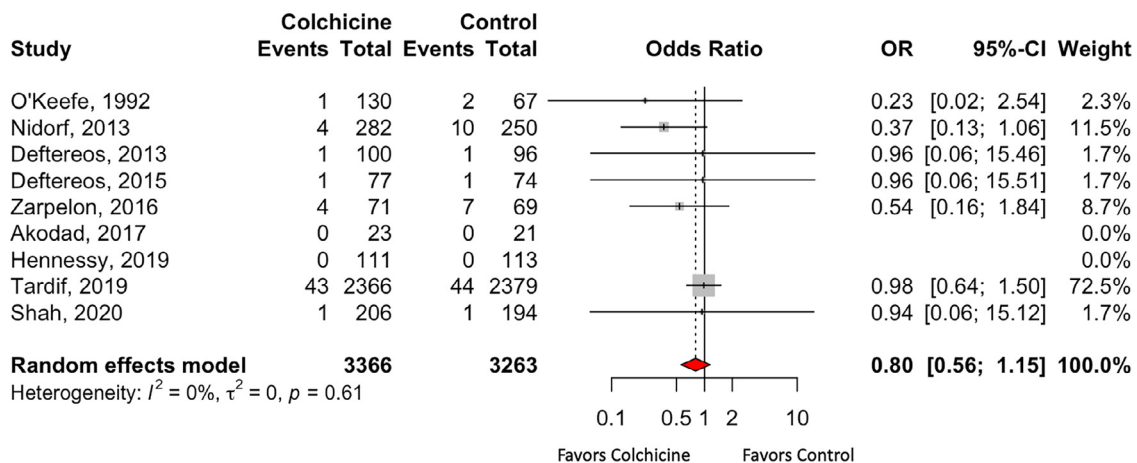


Fig. 3. Effect of colchicine on all-cause mortality.

OR, odds ratio; CI, confidence interval.

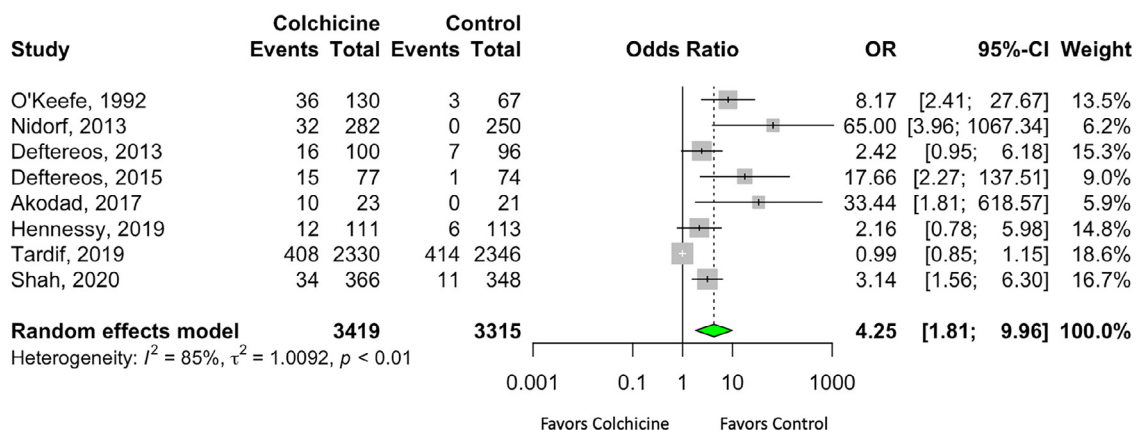


Fig. 4. Effect of colchicine on gastrointestinal events.

OR, odds ratio; CI, confidence interval.

The patients who were treated with colchicine in the included trials had a higher risk of GI events.

Hemkens et al. and Masson et al. had performed meta-analyses to assess the cardiovascular effect of colchicine [22, 23]. Similar to the results of our study, they showed no statistically significant reduction in all-cause mortality with the use of colchicine. This may have resulted from a lack of statistical power owing to the small number of mortality events, because a longer follow-up duration is required to observe mortality events. Regarding the efficacy of colchicine in reducing the incidence of MI in these meta-analyses, the results were controversial. Our result showed a non-significantly lower risk of MI in patients using colchicine, while Hemkens et al. demonstrated a benefit of decreased MI risk under colchicine treatment [22]. However, Masson et al. were not able to find such a benefit [23]. Unlike our study focusing on patients with CAD, these two meta-analyses included trials recruiting patients with other diseases. Some patients included by the study of Hemkens et al. used colchicine for hepatobiliary (mainly cirrhotic) diseases. Some patients included in the study of Masson et al. underwent cardiac surgery due to valvular heart disease. Our meta-analysis excluded these trials on non-CAD patients but included two additional trials on CAD patients; these may lead to different results between their meta-analyses and ours. Besides, these two meta-analyses did not analyze the effect of colchicine on restenosis risk after PCI. However, only a few trials were performed to evaluate such an effect in the past. Our meta-analysis included three such trials and found a significant, protective effect. Moreover, to the best of our knowledge, our meta-analysis is the first to discuss MI and restenosis together.

MI and restenosis are both conditions involving coronary artery occlusion. In our analysis, we found a significant decreasing trend in composite events of MI and restenosis after PCI. Our results also showed that colchicine had significantly lowered the risk of restenosis after PCI. In addition, we found a statistically significant effect of colchicine in reducing the risk of MI within a short period, 6 months, in the subgroup analysis. Considering longer follow-up times, the influence of complex factors on MI outcomes could dilute the effectiveness of a single medication. Therefore, the beneficial effect of colchicine on MI risk could only be detected in short periods in our study. The trial of Tardif et al. had the longest follow-up time of all the trials using colchicine in the acute stage of ACS; thus, the protective effect of colchicine for the risk of MI was low in this study [20]. More human clinical trials are expected to support this mechanism further.

We included three trials reporting restenosis events. The trial of O'Keefe et al. showed a lower protective effect of colchicine on restenosis after PCI than other studies did [10]. However, this trial was performed in 1992 when the standard treatment prior to or after PCI may have been different from today. Besides, in their study, patients only received balloon angioplasty (without stenting), while in the trials by Devereux et al. and Habib et al., patients underwent PCI with bare-metal stents [9, 19]. Compared with balloon angioplasty, stents have a significant advantage in eliminating the elastic recoil and negative remodeling at the treatment site. Both elastic recoil and arterial vessel remodeling are both important mechanisms of restenosis after angioplasty [24, 25]. As colchicine does not show obvious effect on vascular elastic recoil or vessel remodeling, it is less effective in preventing restenosis after plain balloon angioplasty.

Heterogeneity existed in the included trials. For example, each trial had different follow-up durations and used different doses of colchicine. However, the trend in colchicine efficacy was similar. Besides the decreasing trend in total events of MI and restenosis after PCI in the colchicine group, we found consistency in the analysis of all-cause mortality. Our result showed that there was no significant difference in all-cause mortality between the colchicine

and control groups, and this result was consistent in the subgroup analysis. Although the trials of O'Keefe et al., Nidorf et al., and Zarpelon et al. showed trends toward a protective effect of colchicine on mortality, this protective effect did not achieve statistical significance [7, 10, 17]. The patients recruited in these three trials had more stable disease than the patients in other trials, which may highlight the potency of colchicine.

Despite high heterogeneity, our conclusions remained the same in the analysis of GI events and in the sensitivity study. Our results showed a significant increase in GI events with the use of colchicine. Because multiple GI conditions usually manifest simultaneously, we selected the most common GI events for assessment. Two trials (Nidorf et al. and Akodad et al.) had extremely high ORs for GI events, which we believe resulted from their open-label design [7, 18]. Interestingly, in the trial of Tardif et al., the patients in the placebo group experienced more GI events than those in the colchicine group did [20]. Considering that the follow-up time in this trial was almost 2 years, the patients in the placebo group may have taken other types of drug (e.g. antibiotics) for their medical needs, thus causing GI events. However, the authors did report that the colchicine group was indeed at increased risk of nausea and flatulence.

In the subgroup analysis, we defined long duration of follow-up as 6 months or more because patients usually require months of cardiopulmonary rehabilitation after ACS. Usually, physicians recommend cardiac rehabilitation for patients who had an MI within the previous 12 months, and physical rehabilitation is recommended with cardiac monitoring for the first 3 months [26]. We selected 6 months as an adequate cut-off point to estimate the short-term and long-term efficacy of colchicine. We also defined a standard dose of colchicine as 1 mg per day or more. In recent years, clinicians have started to treat patients with low doses of colchicine and 0.5 mg per day is usually defined as a low dose. A colchicine tablet usually contains 0.5 mg or 0.6 mg of active drug, we, therefore, selected at least 1 mg per day as the standard dose for assessment. In some trials, both loading and maintenance doses were used, and we used the loading dose in our analysis due to the short duration of the maintenance dose in these trials (all within 1 month). Use of colchicine was shown to attain statistically significant reduction in restenosis rate after PCI in our study. Trial in the standard-dose group reported events of restenosis and MI after PCI, but trials in the low-dose group only reported MI events. Consequently, meta-analysis for the standard-dose group has a greater statistical power to show the protective effect on MI/ restenosis after PCI than that for the low-dose group.

The strength of our meta-analysis was that we focused on patients with CAD and found a decreasing trend in MI and restenosis after PCI with colchicine use. However, this study has some limitations. First, the event number was low in several reported trials. Fortunately, more than 6000 patients were included in our meta-analysis, and we used a robust method to analyze the data. Second, the studies contained clinical heterogeneity, which we had mentioned in more detail above. Nevertheless, we did find a similar trend of decreasing risk of total events of MI and restenosis after PCI in the colchicine group, and we also observed consistency in the analyses of all-cause mortality and GI events.

Conclusions

The results of this meta-analysis showed that there was a decreasing trend of composite risk of MI and restenosis after PCI with the use of colchicine in patients with CAD. This trend remained significant in the subgroup analyses of long duration of follow-up, short duration of follow-up, and standard colchicine dose. The results also showed a significant decrease in restenosis after PCI with the use of colchicine. However, colchicine did not show a signifi-

cantly beneficial effect on all-cause mortality in CAD patients. During the treatment, GI events should be closely monitored due to the elevated risk of GI side effects. Further prospective large-scale, high-quality studies are required to confirm our findings.

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Disclosures

The authors declare that there is no conflict of interest.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jjcc.2020.11.010](https://doi.org/10.1016/j.jjcc.2020.11.010).

References

- [1] Dalbeth N, Lauterio TJ, Wolfe HR. Mechanism of action of colchicine in the treatment of gout. *Clin Ther* 2014;36:1465–79.
- [2] Imazio M, Brucato A, Trinchero R, Spodick D, Adler Y. Colchicine for pericarditis: hype or hope? *Eur Heart J* 2009;30:532–9.
- [3] Imazio M, Trinchero R, Brucato A, Rovere ME, Gandino A, Cemin R, et al. COLchicine for the Prevention of the Post-pericardiotomy Syndrome (COPPS): a multicentre, randomized, double-blind, placebo-controlled trial. *Eur Heart J* 2010;31:2749–54.
- [4] Imazio M, Brucato A, Ferrazzi P, Pullara A, Adler Y, Barosi A, et al. Colchicine for prevention of postpericardiotomy syndrome and postoperative atrial fibrillation: the COPPS-2 randomized clinical trial. *JAMA* 2014;312:1016–23.
- [5] Inflammation Hansson GK. Inflammation, atherosclerosis, and coronary artery disease. *N Engl J Med* 2005;352:1685–95.
- [6] Singh DP, Mahajan K. Coronary artery thrombus. *StatPearls* [Internet]. Treasure Island, FL: StatPearls Publishing; 2019.
- [7] Nidorf SM, Eikelboom JW, Budgeon CA, Thompson PL. Low-dose colchicine for secondary prevention of cardiovascular disease. *J Am Coll Cardiol* 2013;61:404–10.
- [8] Hennessy T, Soh L, Bowman M, Kurup R, Schultz C, Patel S, et al. The Low Dose Colchicine after Myocardial Infarction (LoDoCo-MI) study: a pilot randomized placebo controlled trial of colchicine following acute myocardial infarction. *Am Heart J* 2019;215:62–9.
- [9] Devereux S, Giannopoulos G, Raisakis K, Kossyvakis C, Kaoukis A, Panagopoulou V, et al. Colchicine treatment for the prevention of bare-metal stent restenosis in diabetic patients. *J Am Coll Cardiol* 2013;61:1679–85.
- [10] O'Keefe JH Jr, McCallister BD, Bateman TM, Kuhnlein DL, Ligon RW, Hartzler GO. Ineffectiveness of colchicine for the prevention of restenosis after coronary angioplasty. *J Am Coll Cardiol* 1992;19:1597–600.
- [11] Imazio M, Gaita F, LeWinter M. Evaluation and treatment of pericarditis: a systematic review. *JAMA* 2015;314:1498–506.
- [12] Christodoulidis G, Vittorio TJ, Fudim M, Lerakis S, Kosmas CE. Inflammation in coronary artery disease. *Cardiol Rev* 2014;22:279–88.
- [13] Thygesen K, Alpert JS, Jaffe AS, Simoons ML, Chaitman BR, White HD, et al. Third universal definition of myocardial infarction. *Eur Heart J* 2012;33:2551–67.
- [14] Higgins JPT, Green S. *Cochrane handbook for systematic reviews of intervention*. Version 5.1.0 [updated March 2011]. The Cochrane Collaboration 2011. Available from www.handbook.cochrane.org.
- [15] Raju NC, Yi Q, Nidorf M, Fagel ND, Hiralal R, Eikelboom JW. Effect of colchicine compared with placebo on high sensitivity C-reactive protein in patients with acute coronary syndrome or acute stroke: a pilot randomized controlled trial. *J Thromb Thrombolysis* 2012;33:88–94.
- [16] Devereux S, Giannopoulos G, Angelidis C, Alexopoulos N, Filippatos G, Papoutsidakis N, et al. Anti-inflammatory treatment with colchicine in acute myocardial infarction: a pilot study. *Circulation* 2015;132:1395–403.
- [17] Zarpelon CS, Netto MC, Jorge JC, Fabris CC, Desengrini D, MdS Jardim, et al. Colchicine to reduce atrial fibrillation in the postoperative period of myocardial revascularization. *Arq Bras Cardiol* 2016;107:4–9.
- [18] Akodad M, Lattuca B, Nagot N, Georgescu V, Buisson M, Cristol JP, et al. COLIN trial: Value of colchicine in the treatment of patients with acute myocardial infarction and inflammatory response. *Arch Cardiovasc Dis* 2017;110:395–402.
- [19] Habib M, Salama I, Al Agha H, Hasira MA. Colchicine treatment for prevention of In Stent Restenosis (CISR Trial). *Am J Emerg Crit Care Med* 2019;2:001–5.
- [20] Tardif JC, Kouz S, Waters DD, Bertrand OF, Diaz R, Maggioni AP, et al. Efficacy and safety of low-dose colchicine after myocardial infarction. *N Engl J Med* 2019;381:2497–505.
- [21] Shah B, Pillinger M, Zhong H, Cronstein B, Xia Y, Lorin JD, et al. Effects of acute colchicine administration prior to percutaneous coronary intervention: COLCHICINE-PCI randomized trial. *Circ Cardiovasc Interv* 2020;13:e008717.
- [22] Hemkens LG, Ewald H, Gloy VL, Arpagaus A, Olu KK, Nidorf M, et al. Cardiovascular effects and safety of long-term colchicine treatment: cochrane review and meta-analysis. *Heart* 2016;102:590–6.
- [23] Masson W, Lobo M, Molinero G, Masson G, Lavalle-Cobo A. Role of colchicine in stroke prevention: An updated meta-analysis. *J Stroke Cerebrovasc Dis* 2020;29:104756.
- [24] Hoffmann R, Mintz G. Coronary in-stent restenosis—predictors, treatment and prevention. *Eur Heart J* 2000;21:1739–49.
- [25] Hoffmann R, Mintz GS, Dussaillant GR, Popma JJ, Pichard AD, Satler LF, et al. Patterns and mechanisms of in-stent restenosis. A serial intravascular ultrasound study. *Circulation* 1996;94:1247–54.
- [26] Servey JT, Stephens M. Cardiac rehabilitation: Improving function and reducing risk. *Am Fam Physician* 2016;94:37–43.